

Impact of untreated dental caries on the daily activities of children

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Abstract

Objectives: Untreated dental caries can compromise oral health-related quality of life (OHRQoL). However, it is not clear which specific daily activities related to OHRQoL are compromised by the presence of this problem. Therefore, the purpose of this study was to identify which specific daily activities are affected by the presence of untreated dental caries using a representative sample of Brazilian children.

Methods: A cross-sectional study using a representative sample of Brazilian children (12 years of age) was conducted. The presence of untreated dental caries was assessed using the “decayed – D” component of the DMFT (decayed, missing, and filled teeth) index. Clinical exams were conducted by trained and calibrated dentists. The daily activities were evaluated using the survey Oral Impacts on Daily Performance (OIDP), considering items and domains. Descriptive, bivariate, and multiple analyses were conducted ($\alpha = 0.05$).

Results: A total of 7,247 children were included, and 39.9 percent had at least one tooth with untreated dental caries, with 33.5 percent experiencing an impact on at least one of the OIDP activities. This study identified a greater chance of the presence of untreated dental caries in children having an impact on the psychological domain ($P = 0.006$), as well as on the “difficulty eating” ($P = 0.007$), and “difficulty sleeping” ($P = 0.023$) items.

Conclusion: The presence of untreated dental caries among Brazilian children was associated with an impact on specific daily activities (difficulty eating and sleeping) and the psychological domain of OHRQoL.

Introduction

Dental caries are a biofilm sugar-dependent disease that can lead to mineral loss of the tooth structure, forming carious lesions, which are considered a signal of disease (1). Furthermore, the disease can silently progress, until changes are clinically visible as white spot lesions, which, if untreated and not controlled, can form dental cavities (1). Although there are strategies for controlling, preventing, and treating carious lesions, part of the population still does not have access to the treatments currently available (2). Some studies have shown a clear decline in the prevalence of dental caries (3-5), but most of them focus only on the experience of caries (4,6,7). This measurement is commonly used (8), reflecting both treated

and untreated caries. However, considering that untreated dental caries represent a real clinical problem that still will be treated and affects a considerable portion of the world's population (9), these lesions need to be better described, especially regarding their impact on daily life and oral health-related quality of life (OHRQoL) (10).

In this way, previous studies have showed that dental caries, as well as their clinical consequences, can reduce OHRQoL, mainly in children (11-14). Untreated dental caries have been associated with specific domains and aspects of quality of life, such as the impact on the overall life of the family (11); social, functional, and emotional limitations of the children (12,13); and specific activities (15).

Moreover, these studies assessed the impact of the untreated dental caries only on overall OHRQoL or in its domains (11-13,16,17). Although there are reports of an impact of these lesions on the specific daily life activities of children (17), those studies were conducted in provincial areas (16,17), or when conducted using a representative sample of a country, they did not assess the chance of occurrence of these lesions considering each daily activity separately (15). In addition, this impact profile can possibly change depending on the characteristics of the country and the prevalence of dental caries in this area. Therefore, in a country with a higher prevalence of dental caries, such as Brazil (≈ 50 percent for children – 12 years old) (18), we expect a different profile of impact (19). In this context, previous study have identified that the psychological and social domains of OHRQoL were associated with untreated caries in Brazilian children (12 years old) (19). However, the specific daily activities of these children that may be affected by the presence of untreated dental caries has not yet been described in the literature using a representative sample of the country. Therefore, the purpose of this study was to identify which specific daily activities are affected by the presence of untreated dental caries using a representative sample of Brazilian children.

Materials and methods

Ethical aspects

The epidemiological survey was conducted respecting the ethical principles of the Resolution of the National Council of Health n° 196/96, according to the Declaration of Helsinki, and was approved and registered by the National Research Ethics Commission (CONEP) under protocol no. 15,498/2010.

Study design, parameters, and the sample used

Data were extracted from databases of the 2010 National Survey of Oral Health Conditions, conducted according to World Health Organization (WHO) criteria (1997) (20,21). Representative samples of the Brazilian population within particular age categories were interviewed and examined by trained and calibrated examiners in their homes regarding their oral health status. After selection through multi-stage probability cluster sampling, with probability proportional to the size and considering a design effect (*deff*) of two, people living in 177 municipalities from the five Brazilian macro-regions (North, Northeast, Central-West, Southeast, and South) were interviewed and examined (21). Professional dentists who were trained and calibrated through the consensus technique conducted the examinations and interviews, with a minimal acceptable Kappa value of 0.65 for each examiner (21). In the present study, were included the children (12 years old) who

were evaluated for dental caries and answered the quality of life questionnaire. This age has been indicated by the WHO as the age of global monitoring of dental caries (20).

Untreated dental caries

The dependent variable – untreated dental caries – was constructed by the DMFT index (decayed, missing, and filled teeth), and only the component “decayed – D” was used to estimate the presence of untreated dental caries. Clinical examinations were performed using a clinical mirror and CPI probe under natural light. A tooth was considered to have decay, as indicated by a cavity, softened tissue at the base, or enamel discoloration, according to WHO criteria (20). For construction of the dependent variables, “without untreated dental caries” was considered those individuals with no affected tooth (without a cavity), and “with untreated dental caries” was considered those individuals with at least one tooth that demonstrated decay (with a cavity). Therefore, missing and filled teeth were not considered in the construction of dependent variables, only the decayed component. Filled teeth with a cavity were considered with untreated dental caries.

Impact on daily life activities

The impact of untreated dental caries on the daily life activities of children was evaluated by the survey Oral Impacts on Daily Performance (OIDP), which measures OHRQoL. The nine daily activities considered in the instrument were difficulty eating, discomfort when brushing, anxiety or irritability, feeling ashamed to smile or talk, difficulty sleeping, effect on leisure, difficulty speaking, disruption of study or work, and an effect on participation in sports. The survey was used in the form of a questionnaire/interview, and each item was preceded by the question, “Some people have problems that might be caused by the teeth. Which of the following situations apply to you in the last six months?” The response options were “no” and “yes” (impact reported) (22). The questions were answered by the children. To evaluate the overall impact on OHRQoL, a dichotomous variable was constructed by attaching nine items, with the response options “no impact” (no impact reported) and “with impact” (impact reported on at least one item). Considering the multidimensional character of quality of life, the nine items of the OIDP were divided among the physical (eating, brushing, speaking, and physical activity), psychological (smiling, sleep, and anxiety), and social (leisure and study) domains, which were evaluated separately and dichotomized in accordance with overall impact (no/yes). Furthermore, each of the nine daily activities (questions) of the OIDP was considered an independent variable in order to assess which specific daily activity could be influenced by the presence of untreated dental caries.

Table 1 Descriptive and Bivariate Analyses of Independent Variables in Relation to the Presence of at Least One Tooth with Untreated Dental Caries among Brazilian Children (12 Years)

Variable	With untreated dental caries				
DEPENDENT	N	%	%	OR	P
Untreated dental caries					
No (0 teeth)	4,220	60.1			
Yes (1 or more teeth)	3,027	39.9	–	–	–
SOCIODEMOGRAPHIC CONDITIONS					
Sex					
Male	2,868	45.7	35.0	1	
Female	4,379	54.3	44.0	1.46	0.001
Self-reported race (skin color)					
White	5,522	31.7	36.8	1	
Asian/Black/Mixed/Indigenous	1,725	68.3	41.3	1.20	0.145
City					
State Capital	3,602	49.7	40.6	1	
Non-capital	3,645	50.3	39.1	0.94	0.522
ORAL HEALTH-RELATED QUALITY LIFE/DAILY ACTIVITIES					
OIDP overall					
No impact	4,693	66.5	33.9	1	
With impact	2,554	33.5	51.8	2.09	<0.001
Physical domain*					
No impact	5,348	73.5	34.8	1	
With impact	1,848	26.5	53.7	2.16	<0.001
Psychological domain*					
No impact	5,517	78.6	35.5	1	
With impact	1,670	21.4	55.7	2.28	<0.001
Social domain*					
No impact	6,601	91.6	37.8	1	
With impact	615	8.4	61.6	2.63	<0.001
OIDP1 – difficulty eating*					
No impact	6,000	82.6	36.6	1	
With impact	1,213	17.4	55.7	2.17	<0.001
OIDP2 – discomfort when brushing*					
No impact	6,318	87.2	37.1	1	
With impact	898	12.8	58.1	2.34	<0.001
OIDP3 – anxiety or irritability*					
No impact	6,447	89.1	37.8	1	
With impact	759	10.9	56.3	2.12	<0.001
OIDP4 – effect on leisure*					
No impact	6,790	94.4	38.5	1	
With impact	429	5.6	62.8	2.70	<0.001
OIDP5 – effect on participation in sports*					
No impact	6,920	95.9	39.0	1	
With impact	301	4.1	59.7	2.32	<0.001
OIDP6 – difficulty speaking*					
No impact	6,891	95.0	39.4	1	
With impact	326	5.0	48.3	1.43	0.153
OIDP7 – Shame to smile or talk*					
No impact	6,235	87.8	38.1	1	
With impact	978	12.2	52.5	1.79	<0.001
OIDP8 – disruption of study*					
No impact	6,852	95.3	38.7	1	
With impact	368	4.7	61.3	2.50	0.004
OIDP9 – difficulty sleeping*					
No impact	6,586	91.9	37.4	1	
With impact	630	8.1	66.8	3.36	<0.001

OIDP, Oral Impacts on Daily Performance; OR, odds ratio.

n = 7247.

*Loss of information.

Independent variables

To control the factors associated with sociodemographic characteristics, the following were also considered independent variables: sex (male/female), skin color (white, Asian descent, black/mixed/indigenous), and city location (capital/not capital).

Statistical analysis

In the data analysis, we used the SPSS Statistics 18.0 software. As the study involved complex cluster sampling, the correction was made via the effect of the sample design, taking into account the effect of clusters and assigning weights to the sampled elements. For categorical variables, the descriptive analysis included the distribution of the sample and corrected relative frequency (%). For the quantitative variables, the average and the standard error (SE) were estimated. To check the association between the presence of untreated dental caries and the impact on the daily life activities, bivariate, and multiple analyses were performed. In the bivariate analysis, odds ratios (OR) and 95% confidence intervals (CI 95%) were estimated. The independent variables with a significance level of less than or equal to 20 percent ($P \leq 0.20$) were selected for multiple analyses. The multiple analyses conducted involved the logistic regression model, with the level of significance adjusted to 5 percent ($P \leq 0.05$) for the variables that remained associated with the outcome (untreated dental caries).

Results

In this study, 7,247 children who met the inclusion criteria were included. Of these, 3,027 (39.9 percent) had at least one tooth with untreated dental caries. The average number of teeth with caries was 1.13 (SE 0.08). Regarding OHRQoL, 33.5 percent of the children experienced an impact on at least one of the daily activities considered in the OIDP. The average number of impacted daily activities was 0.81 (SE 0.06). In addition, the physical domain of the OIDP had the highest prevalence of impact (26.5 percent). In addition, mastication (17.4 percent) was the daily activity most affected. In the bivariate analysis, the overall OIDP, all domains, and daily activities were associated ($P \leq 0.20$) with untreated dental caries (Table 1).

Regarding the analysis of the proportion of children with or without untreated dental caries (at least one tooth decayed) and impact on daily activity, it was identified that a higher proportion of children with untreated caries who were impacted was not observed only in “difficulty in speaking” (Figure 1). In the multiple analyses of factors associated ($P \leq 0.05$) with the presence of untreated dental caries, daily activities that were impacted were “eating/chewing” ($P = 0.007$) and “sleep” ($P = 0.023$). In addition, there is a greater chance of the presence of untreated dental caries in female children ($P = 0.001$), as well as an increased likelihood

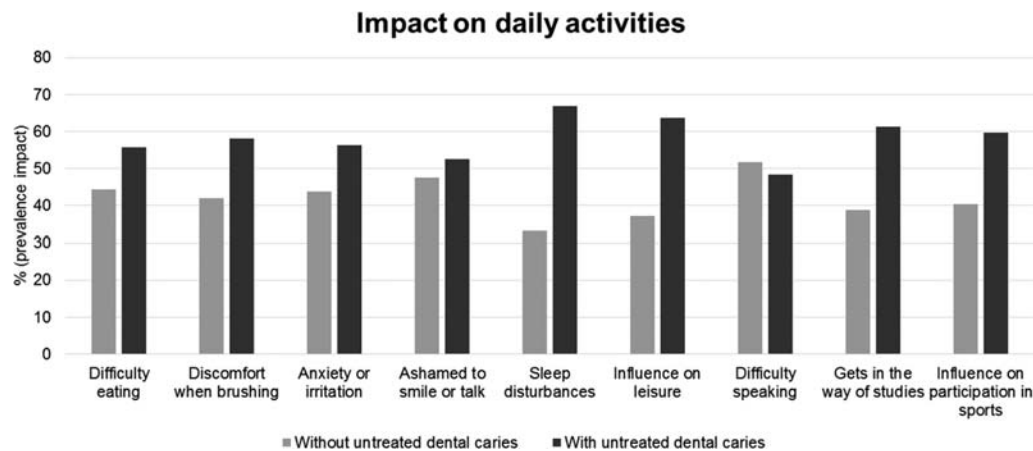


Figure 1 Prevalence (%) of the presence of teeth with and without untreated dental caries within each daily activity among Brazilian children.

of an impact on the psychological domain ($P = 0.006$) (Table 2).

Discussion

A high prevalence of Brazilian children with at least one tooth having untreated dental caries (39.9 percent) was identified. Furthermore, an association of the presence of these lesions with an impact on specific daily activities was found, showing a greater impact of untreated dental caries on the psychological domain and in chewing/eating and sleep activities. Therefore, the study found that the presence of untreated dental caries can affect the execution of specific daily activities in children using a representative sample of children from a developing country. Although previous studies indicate that untreated caries can affect specific daily activities in children (15,16), they do not consider each activity as an independent variable. Other studies using this approach have only suggested specific activities are affected based on activities with higher prevalence of impact. Thus, the present study seems to be the first to explain which specific daily activities are affected among 12-year-old children using a representative sample of a country in development.

Regarding the overall OHRQoL, 33.5 percent of the children experienced an impact on at least one of the daily activities considered in the OIDP. In relation to the associated domain, the association with the psychological domain had already been identified in a previous study (19). However, the present study also identified a greater chance of the presence of untreated dental caries among children having an impact on sleep, one of the daily activities of the psychological domain, characterizing the impact. The pain caused by the presence of untreated dental caries (23), and consequently discomfort/irritation, probably explains the association between untreated dental caries and the psychological domain and sleep. A greater extent and progression of these

lesions can possibly lead to more serious consequences, increasing their impact (11).

A higher chance of untreated dental caries was found among children with an impact on chewing/eating. This impact has not been well explained in the literature. Nevertheless, it has been reported that the presence of this type of lesion can affect nutritional factors, leading to weight loss (24). Moreover, the association between the presence and severity of carious lesions has been associated with severe malocclusion; this fact can be explained by a greater biofilm accumulation in undercuts caused by malocclusion (25). Considering the possibility of malocclusion adversely affecting chewing when concomitantly present with the untreated carious lesion, the identified association may be explained. Note also that this mineral loss in untreated lesions can result in the accumulation of food in the undercuts, along with occlusal changes in the antagonist's tooth, which can impair

Table 2 Multiple Analyses of Variables Associated ($P \leq 0.05$) with Untreated Dental Caries among Brazilian Children

Variable	OR	95% CI	P
SOCIODEMOGRAPHIC CONDITIONS			
Sex			
Male	1		
Female	1.47	1.16-1.85	0.001
ORAL HEALTH-RELATED QUALITY LIFE			
Psychological domain			
No impact	1		
With impact	1.52	1.12-2.05	0.006
OIDP1 – difficulty eating			
No impact	1		
With impact	1.48	1.11-1.99	0.007
OIDP9 – difficulty sleeping			
No impact	1		
With impact	1.86	1.09-3.18	0.023

95% CI, 95% confidence interval; OIDP, Oral Impacts on Daily Performance; OR, odds ratio.

chewing food. In addition, the presence of cavities can cause dental pain, limiting oral functions such as chewing/eating.

In addition to the association with daily activities of children, we identified a greater number of untreated dental caries among girls. A higher prevalence of dental caries among girls has been previously registered (26-28), including among Brazilian children (28). In the present study, variables related to sociodemographic characteristics were included only to control for these conditions, allowing for the identification of other associations, independent of these factors.

The association of untreated dental caries with OHRQoL in children has been established in the literature (11,13,17). Separate evaluation of each daily activity item allows for a better understanding and specification of that activity/function being compromised by the presence of these lesions. From the knowledge of this profile and considering the importance of the normal execution of daily activities, clinical treatment of these lesions should be considered. This treatment could reduce the impact on the daily activities identified. However, it is emphasized that untreated dental caries (cavities) are one of the clinical manifestations of a disease determined by the presence of biofilm and fermentable carbohydrates intake (1). Therefore, treatment of the disease should be considered rather than only its clinical sequela.

In this context, the use of fluoride by several vehicles (29,30), diet control, and preventive educational programs are important approaches to reduce the prevalence of dental caries (1). However, in case of cavity presence, the treatment of lesions should evaluate factors such as the size, depth, and activity of the carious lesion and the caries risk status of the patient. If the lesions have progressed, the cavitated lesions should be treated. Thus, when necessary, restoration treatment should be performed to re-establish, directly or indirectly, the mechanical, aesthetic, and biological function of a natural tooth.

Regarding the limitations of the study, as a cross-sectional and association study, it is not possible to determine whether the presence of untreated caries led to impacts, only to measure the chance of occurrence and identify the associations. In addition, the severity and extent of the lesions were not evaluated.

In conclusion, we identified that the presence of untreated dental caries was associated with an impact on the chewing/eating, sleep, and psychological domains of the OHRQoL of Brazilian children.

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